

INSECTS

Insects first appeared on land more than 400 million years ago and today they account for more species than any other class on the planet.

Insects have evolved diverse lifestyles and although most are terrestrial, there are also numerous freshwater species but almost none are marine. They have a number of key features, such as a small size, an efficient nervous system, high reproductive rates, and – in many cases – the power of flight, which have led to their success.

Insects include (among others) beetles, flies, butterflies, moths, ants, bees, and true bugs. Yet for all their diversity, insects are remarkably similar. Evolution has modified the basic insect anatomy many times over to produce a multitude

of variants based on three major body regions – the head, thorax, and abdomen. The head, made up of six fused segments, houses the brain and carries the major sensory organs: compound eyes, secondary light-receptive organs called ocelli, and the antennae. The mouthparts are modified according to diet, allowing the sucking of liquids or the chewing of solid foods.

The thorax is made up of three segments, each bearing a pair of legs. The posterior two thoracic segments usually each bear a pair of wings. The legs, which are each made up of a number of segments, can be greatly modified to serve a variety of functions from walking and running to jumping, digging, or swimming. The abdomen, which is usually made up of 11 segments, contains the digestive and reproductive organs.

PHYLUM	ARTHROPODA
CLASS	INSECTA
ORDERS	29
FAMILIES	About 1,000
SPECIES	About 1.1 million

DEBATE

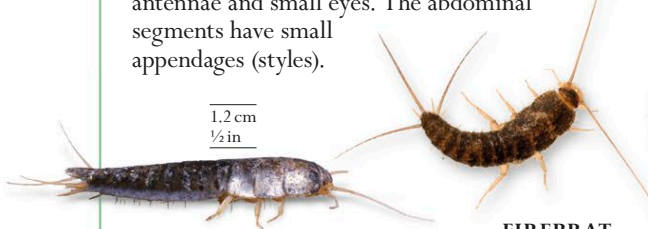
HOW MANY SPECIES?

The actual number of insect species likely to exist far exceeds the 1.1 million so far described, and many more are discovered each year. However, research based on sampling in species-rich rainforests suggests that there could be as many as 10–12 million species of insects.

SILVERFISH

These primitive wingless insects in the order Zygentoma have a lengthy body that is covered with scales. The head has a pair of long antennae and small eyes. The abdominal segments have small appendages (styles).

1.2 cm
½ in



SILVERFISH
Lepisma saccharina
F: Lepismatidae

This common domestic species can sometimes be a nuisance in kitchens, where it feeds on tiny scraps of dropped food.

FIREBRAT
Thermobia domestica
F: Lepismatidae

This insect is found worldwide and lives under stones and in leaf litter. Indoors, it favours warm conditions and can be a pest in bakeries.

1–1.5 cm
⅜–½ in

MAYFLIES

Ephemeroptera are soft-bodied insects with slender legs and two pairs of wings. The head has a pair of short antennae and large, compound eyes. The end of the abdomen has two or three long tail filaments. The life cycle is dominated by the aquatic nymphal stages – the non-feeding adults live for only a few hours or days.

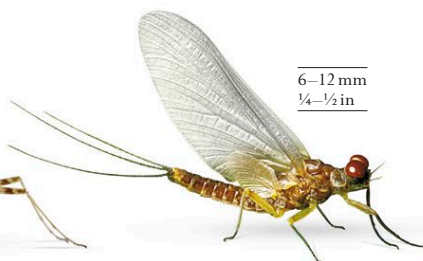
5–7 mm
⅜–⅓ in



POND OLIVE
Cloeon dipterum
F: Baetidae

A widespread European species, this mayfly breeds in a range of habitats from ponds and ditches to troughs and water butts.

6–12 mm
¼–½ in



BLUE-WINGED OLIVE
Ephemera ignita
F: Ephemerellidae

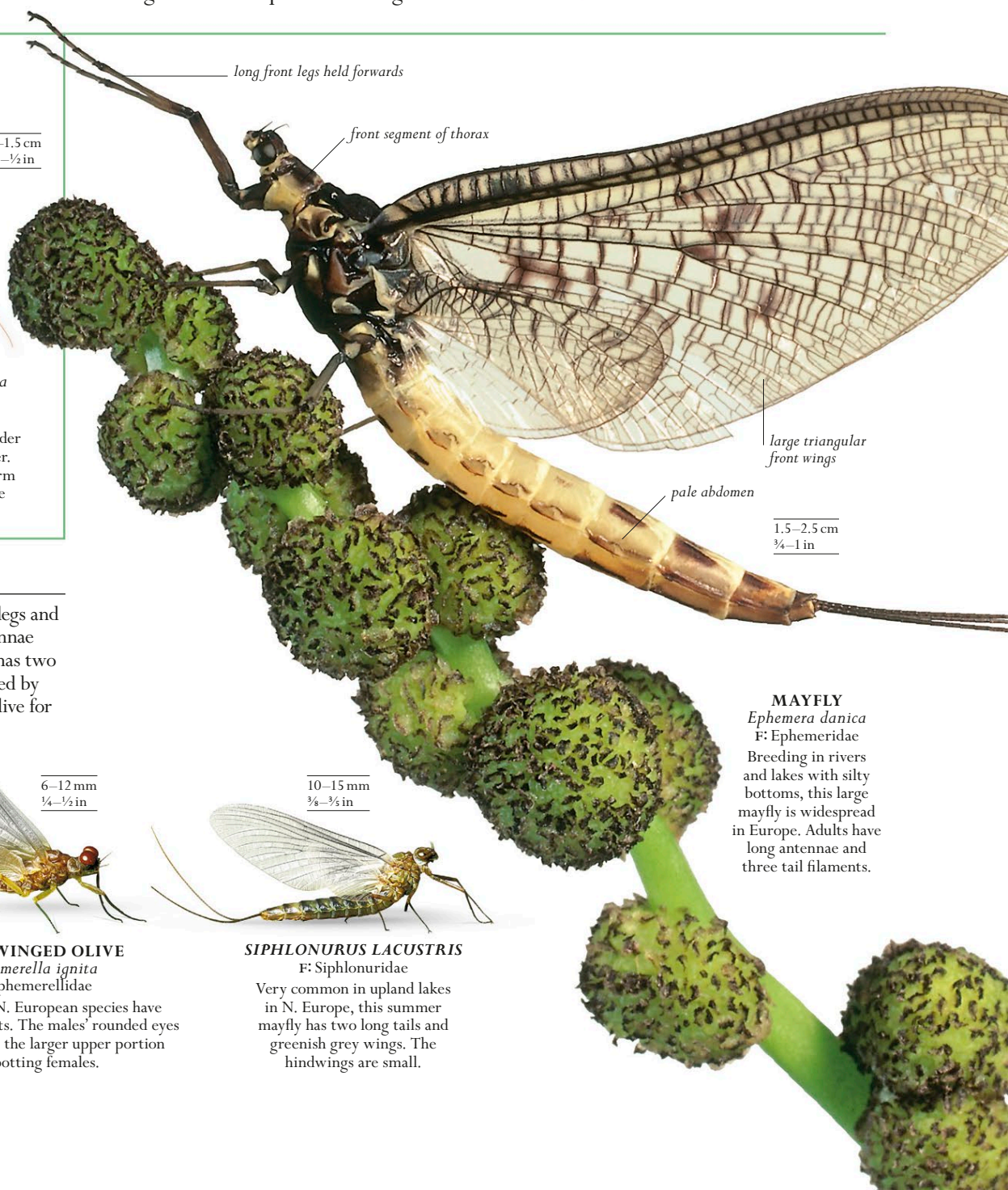
Adults of this N. European species have three tail filaments. The males' rounded eyes have two parts, the larger upper portion for spotting females.

10–15 mm
⅜–⅓ in



SIPHONURUS LACUSTRIS
F: Siphonuridae

Very common in upland lakes in N. Europe, this summer mayfly has two long tails and greenish grey wings. The hindwings are small.



long front legs held forwards

front segment of thorax

large triangular front wings

pale abdomen

1.5–2.5 cm
¾–1 in

MAYFLY
Ephemera danica
F: Ephemeridae
Breeding in rivers and lakes with silty bottoms, this large mayfly is widespread in Europe. Adults have long antennae and three tail filaments.